

Platform Integral as a Cross-System Mechanism for Window Formation in Hybrid Mode Systems

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Abstract

Stable transmission or propagation windows are widely observed in hybrid mode systems, including polaritonic and photonic crystal structures. Conventional interpretations attribute these windows to local extrema, such as minimum residue or maximum group index. However, such pointwise

descriptions do not explain variations in window width or consistency across systems.

In this work, we propose a platform integral framework for window formation. Instead of relying on local extrema, we introduce a global measure based on the structure of system-dependent landscapes:

$$P^* = \int \exp(-X/\langle X \rangle) \cdot \exp(-|\nabla X|/\langle |\nabla X| \rangle) d\Omega$$

where X represents a system-specific variable, such as residue Z_R in polariton systems or group velocity v_g in photonic crystal waveguides.

Across systems, we observe a consistent principle: local extrema determine the central position of the window, while the integrated weight of low-value plateaus determines the usable window width.

This principle is supported by data-driven analysis in polariton systems and consistent with observations in

slow-light photonic crystal systems exhibiting flatband behavior.

These results suggest a cross-system structural mechanism for window formation based on platform integrals, providing a unified perspective beyond extremum-based interpretations.

1. Introduction

Stable windows of transmission or propagation arise in a variety of hybrid mode systems. Examples include polaritonic structures and photonic crystal waveguides.

Traditional approaches characterize these windows using local extrema of system parameters, such as minimum loss or maximum group index. While these quantities indicate the location of a mode, they do not fully capture the extent or usability of the window.

This motivates the need for a structural description that goes beyond pointwise extrema.

2. Platform Integral Framework

We introduce a global measure defined over a system-dependent landscape X :

$$P^* = \int e^{\{-X/\langle X \rangle\}} \cdot e^{\{-|\nabla X|/\langle |\nabla X| \rangle\}} d\Omega$$

The first term emphasizes low-value regions, while the second term favors flatness.

The integral captures the combined effect of magnitude and spatial smoothness, providing a measure of plateau structure.

3. Cross-System Mapping

The framework applies to different systems through appropriate identification of X :

System	Landscape Variable X	Observable Window
Polariton Residue	$Z_R(\theta)$	quasi-flat / protected window
Photonic crystal	Group velocity $v_g(\lambda)$	slow-light window

In both cases, the observable window corresponds to regions where X is low and slowly varying.

4. Core Principle

We summarize the structural rule as:

Local extrema determine the central position of a window, while the integrated weight of low-value plateaus determines the usable window width.

This replaces a purely local description with a global structural mechanism.

5. Empirical Support

In polariton systems, analysis of Hopfield coefficient data shows that the platform integral P^* correlates with observed window width.

In photonic crystal systems, flatband slow-light modes exhibit extended regions of low and nearly constant group velocity, consistent with a large platform integral.

These observations support the proposed mechanism across systems.

6. Conclusion

Window formation in hybrid mode systems is governed by the global structure of the underlying landscape rather than local extrema alone.

The platform integral provides a minimal and generalizable framework for describing this behavior, suggesting a unified structural mechanism across different physical systems.

Keywords

hybrid modes, platform integral, residue landscape, slow light, photonic crystal, polariton, flatband, window formation, mode structure, cross-system mechanism
